## NMC 2012 PRIMARY LEVEL PAPER 2 QUESTIONS

### SECTION A

**Qn. 1.** Arrange the following seven numbers from the smallest to the largest: 1, 0.5, $\frac{1}{3}$, $-2$, $\frac{3}{4}$, 0.8, $-9$

**Qn. 2.** For each of the following measures, indicate the larger one:

(a) 9001 grams or 9 kilograms

(b) 4 metres or 3.999 millimetres

(c) $\frac{4}{3}$ litres or $\frac{3}{2}$ litres

**Qn. 3.** Juliet multiplies a number $x$ by 3, and then divides the result by 5 and she gets $\frac{3}{10}$. What is the value of $x$?

**Qn. 4.** Find digits $a, b, c, d, e$ in this addition:

$$\begin{array}{cccccc} & a & 3 & b & 3 & 4 \\ + & 3 & 4 & 7 & c & d \\ \hline & 8 & e & 3 & 3 & 7 \end{array}$$

**Qn. 5.** A meat pie was cut into 32 pieces. Each of the two children ate 2 pieces of the pie. Each adult ate 4 pieces of the pie. How many people were there?

**Qn. 6.** The size of angle MBD is _____

**Qn. 7.** A choir of 124 girls is to be transported in minibuses. Each bus can carry 8 girls only. What is the smallest possible number of buses that are needed to transport all the 124 girls?

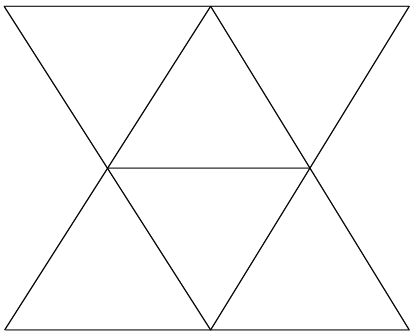
**Qn. 8.** There are 4 more chickens than rabbits on a farm. The total number of legs (of chickens and rabbits) is equal to 44. How many chickens and rabbits are there on the farm?

**Qn. 9.** The shaded shape is made up of 5 congruent squares. The side of one square is 4cm. Find the area of the shaded shape.

**Qn. 10.** Four congruent isosceles right triangles are cut from the four corners of a square with a side of 20cm. The length of one side of the triangles is equal to 4cm as shown on the diagram. What is the area of the remaining octagon?

### SECTION B

**Qn. 11.** Count all the triangles in this figure:



**Qn. 12.** In a certain primary school, 50% of the pupils are younger than 10 years, $\frac{1}{20}$ are 10 years old and $\frac{1}{10}$ are older than 10 years but younger than 12 years, the remaining 70 pupils are 12 years and above. How many pupils are 10 years old?

**Qn. 13.** Joan drives from town $A$ to town $B$. Town $A$ is 1000km from town $B$. She drives the first half of the journey at 100km per hour. Feeling thirsty, she makes a stop over at a restaurant for 35 minutes. She then drives the rest of the journey at 80km per hour.

(a) How long did it take Joan to complete the journey?

(b) If she had driven non-stop at 100km per hour, how long would the journey have taken?

**Qn. 14.** The following notice is at the entrance to a school library:

**OPEN:**
MONDAY - FRIDAY FROM 8AM - 5PM
SATURDAY FROM 9AM - 2PM

**CLOSED:**
SUNDAY AND PUBLIC HOLIDAYS.

(a) How many hours is the library open each week if the week does not have any public holiday?

(b) How many hours was the library open during the month of April 2012?

**Qn. 15.** Each pupil in Primary seven (P7) was asked to indicate their favourite subject. The following results were obtained:

| Subject | Number of pupils |
|---|---|
| Science | 51 |
| Mathematics | 61 |
| SST | 49 |
| English | 63 |

(a) How many pupils were in the P7 class?

(b) What percentage of pupils did not choose Mathematics as their favourite subject?

(c) In this school, there were 4 streams for P7 with each stream having an equal number of pupils. How many pupils were in each stream?

## NMC 2012 PRIMARY LEVEL PAPER 2 SOLUTIONS

### SECTION A : 5 marks each

**Qn1.** $-9, -2, \frac{1}{3}, \frac{1}{2}, \frac{3}{4}, 0.8, 1.$

**Qn2.** (a) 9001 grams.

(b) 4 metres.

(c) $\frac{3}{2}$ metres.

**Qn3.** We are given that $\frac{3}{5} = \frac{3}{10}$. If we multiply both sides by $\frac{5}{3}$ we get $x = \frac{3}{10} \times \frac{5}{3} = \frac{5}{10} = \frac{1}{2}$.

**Qn4.**

| | | | | |
|---|---|---|---|---|
| a | 3 | b | 3 | 4 |
| 3 | 4 | 7 | c | d |
| 8 | e | 3 | 3 | 7 |

$4 + d = 7$, so $d = 7 - 4 = 3$. $3 + c = c$, so $c = 3 - 3 = 0$. $b + 7 = 13$, so $b = 13 - 7 = 6$. To get $e$ note that I carried 1, so $e = 4 + 4 = 8$. Lastly, $a + 3 = 8$, so $a = 8 - 3 = 5$.

**Qn5.** The two kids together ate $2 \times 2 = 4$ pieces. The number of pieces remaining for the adults are $32 - 4 = 28$. Each adult ate 4 pieces, so there were $\frac{28}{4} = 7$ adults.

**Qn6.** $\angle ACM + \angle CAM + \angle AMC = 180$. Therefore, $\angle AMC = 180 - 56 - 78 = 46^o$. $\angle DMB = \angle AMC$ because they are alternate angles.

Since the all angles $\triangle DBM$ add up to $180^o$ we have $\angle DMB + 62 + \angle DMB = 180$. Or $\angle DBM = 180 - 62 - \angle DMB = 180 - 62 - 46 = 72^o$.,

**Qn7.** $\frac{124}{8} = 15.5$ Since we cannot have 0.5 of a van, the smallest number of vans is 16.

**Qn8.** Let $x$ be the number of rabbits. This means that there are $x + 4$ chicken. Each rabbit has four legs meaning that the $x$ rabbits have $4x$ legs. Each chicken has 2 legs meaning that all chicken have $2(x+)4$ legs. The total number of legs being $4x + 2(x + 4) = 4x + 2x + 8 = 6x + 8$. We solve $6x + 8 = 44$ or $6x = 44 - 8 = 36$ or $x = \frac{36}{6} = 6$. Therefore, there are 6 rabbits and $6 + 4 = 10$ chickens.

**Qn9.** Each square has area $A = \text{side}^2 = (4cm)^2 = 16cm^2$. The area of the shaded shape is equal to $5 \times A = 5 \times 16cm^2 = 80cm^2$.

**Qn10.**

$$\begin{aligned}\text{Area of the remaining Octagon} &= \text{Area of the square} - \text{Area of the four triangles}\\ &= \text{side}^2 - 4 \times \frac{1}{2}bh\\ &= (20cm)^2 - 4 \times \frac{1}{2} \times 4cm \times 4cm\\ &= 400cm^2 - 2 \times 4cm \times 4cm = 400cm^2 - 32cm^2 = 368cm^2.\end{aligned}$$

## SECTION B: 10 MARKS EACH

**Qn11.** There are 8 triangles.

**Qn12.** $50\% = \frac{50}{100} = \frac{1}{2}$. So $\frac{1}{2}$ of the pupils are younger than 10 years, $\frac{1}{20}$ are 10 years old and $\frac{1}{10}$ are older than 10 years but younger than 12 years. Let $x$ be the total number of pupils. This means that pupils who are younger than 12 years are $\frac{1}{2}x + \frac{1}{20}x + \frac{1}{10}x = \frac{10+2+1}{20}x = \frac{13}{20}x$. Those who are 12 or older than 12 years are $x - \frac{13}{20}x = \frac{7}{20}x$. We are given that those who are 12 years and above are 70, so we solve $\frac{7}{20}x = 70$, meaning that $x = 70 \times \frac{20}{7} = 200$. Therefore, the number of pupils who are 10 years old is $\frac{1}{10}x = \frac{1}{10} \times 200 = 20$.

**Qn13.** (a) We use the formula Speed $= \dfrac{\text{Distance Covered}}{\text{Time taken to cover the distance}}$ or

Time taken $= \dfrac{\text{Distance Covered}}{\text{Speed}}$.

The time taken to cover half of the distance is $\frac{500}{100} = 5$ hours. The time taken to cover the other half is $\frac{500}{80} = 6.25$ hours= 6 hours and 15 minutes. Total time taken including the resting time is $5hrs + 6hrs + 15mins + 35mins = 12hrs$ and 50 minutes which equals 11.83 hours.

(b) If she drove non-stop at $100km$ per hour, she would have taken $\frac{1000}{100} = 10$ hours.

**Qn14.** (a) It is open 9 hours for 5 days, which in total comes to $9 \times 5 = 45$ hours. On Saturday, it is open for 5 hours. So, if there are no public holidays, the library is open for $45 + 5 = 50$ hours in a week.

(b) April has 30 days, 5 of which were Sundays, 2 were public holidays(Good Friday and Easter Monday) and 4 were Saturdays. So, the library was open for only 19 weekdays, for a total of $19 \times 9 = 171$ hours. For all the 4 Saturdays, the library was opened for $4 \times 5 = 20$ hours. So, in April, the library was opened for $171 + 20 = 191$ hours.

**Qn15.** (a) There were $51 + 61 + 49 + 63 = 224$ pupils in the $P7$ class.

(b) Those who did not choose Mathematics as their favorite subject are $51 + 49 + 63 = 163$. The percentage is $\frac{163}{224} \times 100\% = 72.8\%$.

(c) Each stream has $\frac{224}{4} = 56$ Pupils.